MA1522 Linear Algebra for Computing
L2 Lesson 21



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Question 1

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Consider the following linear system:

$$\begin{cases} ax + y + z = a^3 \\ x + ay + z = 1 \\ x + y + az = a, \end{cases}$$

where a is a real constant. Find the conditions on a such that:

a. the system has no solution;

b. the system has a unique solution;

c. the system has infinitely many solutions.

[Pg 76–77 of Chapter 1.]

Question 1: Solution

$$\left(\begin{array}{ccc|c} a & 1 & 1 & a^3 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & a \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|c} 1 & a & 1 & 1 \\ a & 1 & 1 & a^3 \\ 1 & 1 & a & a \end{array}\right) \xrightarrow[R_3 - R_1]{R_2 - aR_1} \left(\begin{array}{ccc|c} 1 & a & 1 & 1 \\ 0 & 1-a^2 & 1-a & a^3-a \\ 0 & 1-a & a-1 & a-1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & a & 1 & 1 \\ 0 & 1-a & a-1 & a-1 \\ 0 & 1-a^2 & 1-a & a^3-a \end{array}\right) \xrightarrow{R_3 - (1+a)R_2} \left(\begin{array}{ccc|c} 1 & a & 1 & 1 \\ 0 & 1-a & a-1 & a-1 \\ 0 & 0 & (1-a)(2+a) & (1-a)^2(1+a) \end{array}\right)$$

a. If $a = -2$, the matrix is reduced to $\left(\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & 3 & -3 & -3 \\ 0 & 0 & 0 & 9 \end{array}\right)$, and the linear system has no solution.

b. If $a \neq 1$ and $a \neq -2$, then linear the system has a unique solution.

c. If $a = 1$, the matrix is reduced to $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$, and the linear system has infinitely many solutions (with 2 arbitrary parameters).

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Question 2

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Let a and b be real constants, and let

$$\boldsymbol{A} = \begin{pmatrix} 2 & -4 & 4 & -2 \\ 0 & a & -5 & a \\ 0 & -b & 10 & -b \end{pmatrix}.$$

a. Find the conditions on a and b for $\boldsymbol{A}$ to have LU-factorization.

b. Find the LU-factorization of $\boldsymbol{A}$ (in terms of a and/or b if necessary) when it exists.

[Pg 125 of Chapter 2.]

Question 2: Solution

$$\boldsymbol{A} = \begin{pmatrix} 2 & -4 & 4 & -2 \\ 0 & a & -5 & a \\ 0 & -b & 10 & -b \end{pmatrix}.$$

If $a = 0$, then $\boldsymbol{A}$ does not have LU-decomposition unless $b = 0$.

When $a = 0 = b$, $\begin{pmatrix} 2 & -4 & 4 & -2 \\ 0 & 0 & -5 & 0 \\ 0 & 0 & 10 & 0 \end{pmatrix} \xrightarrow{R_3+2R_2} \begin{pmatrix} 2 & -4 & 4 & -2 \\ 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$.

Thus, $\boldsymbol{A} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix} \begin{pmatrix} 2 & -4 & 4 & -2 \\ 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ is the LU-factorization of $\boldsymbol{A}$.

If $a \neq 0$, then: $\begin{pmatrix} 2 & -4 & 4 & -2 \\ 0 & a & -5 & a \\ 0 & -b & 10 & -b \end{pmatrix} \xrightarrow{R_3+\frac{b}{a}R_2} \begin{pmatrix} 2 & -4 & 4 & -2 \\ 0 & a & -5 & a \\ 0 & 0 & -\frac{5b}{a}+10 & 0 \end{pmatrix}$.

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Question 3

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Let c be a real constant, and let

$$\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & -1 & c \\ 1 & 4 & 1 & c^2 \\ 1 & 8 & -1 & c^3 \end{pmatrix}.$$

Determine the values of c for which $\boldsymbol{A}$ is invertible. [Pg 151 of Chapter 2.]

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Solution.

$$\boldsymbol{A} \xrightarrow[R_4-R_1]{\substack{R_2-R_1 \\ R_3-R_1}} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & c-1 \\ 0 & 3 & 0 & c^2-1 \\ 0 & 7 & -2 & c^3-1 \end{pmatrix} \xrightarrow[R_4-7R_2]{R_3-3R_2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & c-1 \\ 0 & 0 & 6 & c^2-3c+2 \\ 0 & 0 & 12 & c^3-7c+6 \end{pmatrix}$$

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Thus $\boldsymbol{A}$ is invertible

$\Leftrightarrow \det(\boldsymbol{A}) \neq 0 \Leftrightarrow c^3 - 2c^2 - c + 2 = (c+1)(c-1)(c-2) \neq 0 \Leftrightarrow c \neq -1, 1, 2.$

□

Next Lesson ...

We will revise **Chapters 3 and 4**.